

Internet Engineering Task Force (IETF)
Request for Comments: 6347
Obsoletes: 4347
Category: Standards Track
ISSN: 2070-1721

E. Rescorla
RTFM, Inc.
N. Modadugu
Google, Inc.
January 2012

Datagram Transport Layer Security Version 1.2

Abstract

This document specifies version 1.2 of the Datagram Transport Layer Security (DTLS) protocol. The DTLS protocol provides communications privacy for datagram protocols. The protocol allows client/server applications to communicate in a way that is designed to prevent eavesdropping, tampering, or message forgery. The DTLS protocol is based on the Transport Layer Security (TLS) protocol and provides equivalent security guarantees. Datagram semantics of the underlying transport are preserved by the DTLS protocol. This document updates DTLS 1.0 to work with TLS version 1.2.

Status of This Memo

This is an Internet Standards Track document.

This document is a product of the Internet Engineering Task Force (IETF). It represents the consensus of the IETF community. It has received public review and has been approved for publication by the Internet Engineering Steering Group (IESG). Further information on Internet Standards is available in Section 2 of RFC 5741.

Information about the current status of this document, any errata, and how to provide feedback on it may be obtained at <http://www.rfc-editor.org/info/rfc6347>.

Copyright Notice

Copyright (c) 2012 IETF Trust and the persons identified as the document authors. All rights reserved.

This document is subject to BCP 78 and the IETF Trust's Legal Provisions Relating to IETF Documents (<http://trustee.ietf.org/license-info>) in effect on the date of publication of this document. Please review these documents carefully, as they describe your rights and restrictions with respect to this document. Code Components extracted from this document must include Simplified BSD License text as described in Section 4.e of the Trust Legal Provisions and are provided without warranty as described in the Simplified BSD License.

This document may contain material from IETF Documents or IETF Contributions published or made publicly available before November 10, 2008. The person(s) controlling the copyright in some of this material may not have granted the IETF Trust the right to allow modifications of such material outside the IETF Standards Process. Without obtaining an adequate license from the person(s) controlling the copyright in such materials, this document may not be modified outside the IETF Standards Process, and derivative works of it may not be created outside the IETF Standards Process, except to format it for publication as an RFC or to translate it into languages other than English.

Table of Contents

1. Introduction	4
1.1. Requirements Terminology	5
2. Usage Model	5
3. Overview of DTLS	5
3.1. Loss-Insensitive Messaging	6
3.2. Providing Reliability for Handshake	6
3.2.1. Packet Loss	6
3.2.2. Reordering	7
3.2.3. Message Size	7
3.3. Replay Detection	7
4. Differences from TLS	7
4.1. Record Layer	8
4.1.1. Transport Layer Mapping	10
4.1.1.1. PMTU Issues	10
4.1.2. Record Payload Protection	12
4.1.2.1. MAC	12
4.1.2.2. Null or Standard Stream Cipher	13
4.1.2.3. Block Cipher	13
4.1.2.4. AEAD Ciphers	13
4.1.2.5. New Cipher Suites	13
4.1.2.6. Anti-Replay	13
4.1.2.7. Handling Invalid Records	14
4.2. The DTLS Handshake Protocol	14
4.2.1. Denial-of-Service Countermeasures	15
4.2.2. Handshake Message Format	18
4.2.3. Handshake Message Fragmentation and Reassembly	19
4.2.4. Timeout and Retransmission	20
4.2.4.1. Timer Values	24
4.2.5. ChangeCipherSpec	25
4.2.6. CertificateVerify and Finished Messages	25
4.2.7. Alert Messages	25
4.2.8. Establishing New Associations with Existing Parameters	25
4.3. Summary of New Syntax	26
4.3.1. Record Layer	26
4.3.2. Handshake Protocol	27
5. Security Considerations	27
6. Acknowledgments	28
7. IANA Considerations	28
8. Changes since DTLS 1.0	29
9. References	30
9.1. Normative References	30
9.2. Informative References	31

1. Introduction

TLS [TLS] is the most widely deployed protocol for securing network traffic. It is widely used for protecting Web traffic and for e-mail protocols such as IMAP [IMAP] and POP [POP]. The primary advantage of TLS is that it provides a transparent connection-oriented channel. Thus, it is easy to secure an application protocol by inserting TLS between the application layer and the transport layer. However, TLS must run over a reliable transport channel -- typically TCP [TCP]. Therefore, it cannot be used to secure unreliable datagram traffic.

An increasing number of application layer protocols have been designed that use UDP transport. In particular, protocols such as the Session Initiation Protocol (SIP) [SIP] and electronic gaming protocols are increasingly popular. (Note that SIP can run over both TCP and UDP, but that there are situations in which UDP is preferable.) Currently, designers of these applications are faced with a number of unsatisfactory choices. First, they can use IPsec [RFC4301]. However, for a number of reasons detailed in [WHYIPSEC], this is only suitable for some applications. Second, they can design a custom application layer security protocol. Unfortunately, although application layer security protocols generally provide superior security properties (e.g., end-to-end security in the case of S/MIME), they typically require a large amount of effort to design -- in contrast to the relatively small amount of effort required to run the protocol over TLS.

In many cases, the most desirable way to secure client/server applications would be to use TLS; however, the requirement for datagram semantics automatically prohibits use of TLS. This memo describes a protocol for this purpose: Datagram Transport Layer Security (DTLS). DTLS is deliberately designed to be as similar to TLS as possible, both to minimize new security invention and to maximize the amount of code and infrastructure reuse.

DTLS 1.0 [DTLS1] was originally defined as a delta from [TLS11]. This document introduces a new version of DTLS, DTLS 1.2, which is defined as a series of deltas to TLS 1.2 [TLS12]. There is no DTLS 1.1; that version number was skipped in order to harmonize version numbers with TLS. This version also clarifies some confusing points in the DTLS 1.0 specification.

Implementations that speak both DTLS 1.2 and DTLS 1.0 can interoperate with those that speak only DTLS 1.0 (using DTLS 1.0 of course), just as TLS 1.2 implementations can interoperate with previous versions of TLS (see Appendix E.1 of [TLS12] for details), with the exception that there is no DTLS version of SSLv2 or SSLv3, so backward compatibility issues for those protocols do not apply.

1.1. Requirements Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119 [REQ].

2. Usage Model

The DTLS protocol is designed to secure data between communicating applications. It is designed to run in application space, without requiring any kernel modifications.

Datagram transport does not require or provide reliable or in-order delivery of data. The DTLS protocol preserves this property for payload data. Applications such as media streaming, Internet telephony, and online gaming use datagram transport for communication due to the delay-sensitive nature of transported data. The behavior of such applications is unchanged when the DTLS protocol is used to secure communication, since the DTLS protocol does not compensate for lost or re-ordered data traffic.

3. Overview of DTLS

The basic design philosophy of DTLS is to construct "TLS over datagram transport". The reason that TLS cannot be used directly in datagram environments is simply that packets may be lost or reordered. TLS has no internal facilities to handle this kind of unreliability; therefore, TLS implementations break when rehosted on datagram transport. The purpose of DTLS is to make only the minimal changes to TLS required to fix this problem. To the greatest extent possible, DTLS is identical to TLS. Whenever we need to invent new mechanisms, we attempt to do so in such a way that preserves the style of TLS.

Unreliability creates problems for TLS at two levels:

1. TLS does not allow independent decryption of individual records. Because the integrity check depends on the sequence number, if record N is not received, then the integrity check on record N+1 will be based on the wrong sequence number and thus will fail. (Note that prior to TLS 1.1, there was no explicit IV and so decryption would also fail.)
2. The TLS handshake layer assumes that handshake messages are delivered reliably and breaks if those messages are lost.

The rest of this section describes the approach that DTLS uses to solve these problems.

3.1. Loss-Insensitive Messaging

In TLS's traffic encryption layer (called the TLS Record Layer), records are not independent. There are two kinds of inter-record dependency:

1. Cryptographic context (stream cipher key stream) is retained between records.
2. Anti-replay and message reordering protection are provided by a MAC that includes a sequence number, but the sequence numbers are implicit in the records.

DTLS solves the first problem by banning stream ciphers. DTLS solves the second problem by adding explicit sequence numbers.

3.2. Providing Reliability for Handshake

The TLS handshake is a lockstep cryptographic handshake. Messages must be transmitted and received in a defined order; any other order is an error. Clearly, this is incompatible with reordering and message loss. In addition, TLS handshake messages are potentially larger than any given datagram, thus creating the problem of IP fragmentation. DTLS must provide fixes for both of these problems.

3.2.1. Packet Loss

DTLS uses a simple retransmission timer to handle packet loss. The following figure demonstrates the basic concept, using the first phase of the DTLS handshake:

```

Client                               Server
-----
ClientHello                          ----->
                                     X<-- HelloVerifyRequest
                                     (lost)

[Timer Expires]

ClientHello                          ----->
(retransmit)

```

Once the client has transmitted the ClientHello message, it expects to see a HelloVerifyRequest from the server. However, if the server's message is lost, the client knows that either the ClientHello or the HelloVerifyRequest has been lost and retransmits. When the server receives the retransmission, it knows to retransmit.

The server also maintains a retransmission timer and retransmits when that timer expires.

Note that timeout and retransmission do not apply to the HelloVerifyRequest, because this would require creating state on the server. The HelloVerifyRequest is designed to be small enough that it will not itself be fragmented, thus avoiding concerns about interleaving multiple HelloVerifyRequests.

3.2.2. Reordering

In DTLS, each handshake message is assigned a specific sequence number within that handshake. When a peer receives a handshake message, it can quickly determine whether that message is the next message it expects. If it is, then it processes it. If not, it queues it for future handling once all previous messages have been received.

3.2.3. Message Size

TLS and DTLS handshake messages can be quite large (in theory up to $2^{24}-1$ bytes, in practice many kilobytes). By contrast, UDP datagrams are often limited to <1500 bytes if IP fragmentation is not desired. In order to compensate for this limitation, each DTLS handshake message may be fragmented over several DTLS records, each of which is intended to fit in a single IP datagram. Each DTLS handshake message contains both a fragment offset and a fragment length. Thus, a recipient in possession of all bytes of a handshake message can reassemble the original unfragmented message.

3.3. Replay Detection

DTLS optionally supports record replay detection. The technique used is the same as in IPsec AH/ESP, by maintaining a bitmap window of received records. Records that are too old to fit in the window and records that have previously been received are silently discarded. The replay detection feature is optional, since packet duplication is not always malicious, but can also occur due to routing errors. Applications may conceivably detect duplicate packets and accordingly modify their data transmission strategy.

4. Differences from TLS

As mentioned in Section 3, DTLS is intentionally very similar to TLS. Therefore, instead of presenting DTLS as a new protocol, we present it as a series of deltas from TLS 1.2 [TLS12]. Where we do not explicitly call out differences, DTLS is the same as in [TLS12].

4.1. Record Layer

The DTLS record layer is extremely similar to that of TLS 1.2. The only change is the inclusion of an explicit sequence number in the record. This sequence number allows the recipient to correctly verify the TLS MAC. The DTLS record format is shown below:

```
struct {  
    ContentType type;  
    ProtocolVersion version;  
    uint16 epoch; // New field  
    uint48 sequence_number; // New field  
    uint16 length;  
    opaque fragment[DTLSPlaintext.length];  
} DTLSPlaintext;
```

type

Equivalent to the type field in a TLS 1.2 record.

version

The version of the protocol being employed. This document describes DTLS version 1.2, which uses the version { 254, 253 }. The version value of 254.253 is the 1's complement of DTLS version 1.2. This maximal spacing between TLS and DTLS version numbers ensures that records from the two protocols can be easily distinguished. It should be noted that future on-the-wire version numbers of DTLS are decreasing in value (while the true version number is increasing in value.)

epoch

A counter value that is incremented on every cipher state change.

sequence_number

The sequence number for this record.

length

Identical to the length field in a TLS 1.2 record. As in TLS 1.2, the length should not exceed 2^{14} .

fragment

Identical to the fragment field of a TLS 1.2 record.

DTLS uses an explicit sequence number, rather than an implicit one, carried in the sequence_number field of the record. Sequence numbers are maintained separately for each epoch, with each sequence_number initially being 0 for each epoch. For instance, if a handshake message from epoch 0 is retransmitted, it might have a sequence number after a message from epoch 1, even if the message from epoch 1

was transmitted first. Note that some care needs to be taken during the handshake to ensure that retransmitted messages use the right epoch and keying material.

If several handshakes are performed in close succession, there might be multiple records on the wire with the same sequence number but from different cipher states. The epoch field allows recipients to distinguish such packets. The epoch number is initially zero and is incremented each time a ChangeCipherSpec message is sent. In order to ensure that any given sequence/epoch pair is unique, implementations MUST NOT allow the same epoch value to be reused within two times the TCP maximum segment lifetime. In practice, TLS implementations rarely rehandshake; therefore, we do not expect this to be a problem.

Note that because DTLS records may be reordered, a record from epoch 1 may be received after epoch 2 has begun. In general, implementations SHOULD discard packets from earlier epochs, but if packet loss causes noticeable problems they MAY choose to retain keying material from previous epochs for up to the default MSL specified for TCP [TCP] to allow for packet reordering. (Note that the intention here is that implementors use the current guidance from the IETF for MSL, not that they attempt to interrogate the MSL that the system TCP stack is using.) Until the handshake has completed, implementations MUST accept packets from the old epoch.

Conversely, it is possible for records that are protected by the newly negotiated context to be received prior to the completion of a handshake. For instance, the server may send its Finished message and then start transmitting data. Implementations MAY either buffer or discard such packets, though when DTLS is used over reliable transports (e.g., SCTP), they SHOULD be buffered and processed once the handshake completes. Note that TLS's restrictions on when packets may be sent still apply, and the receiver treats the packets as if they were sent in the right order. In particular, it is still impermissible to send data prior to completion of the first handshake.

Note that in the special case of a rehandshake on an existing association, it is safe to process a data packet immediately, even if the ChangeCipherSpec or Finished messages have not yet been received provided that either the rehandshake resumes the existing session or that it uses exactly the same security parameters as the existing association. In any other case, the implementation MUST wait for the receipt of the Finished message to prevent downgrade attack.

As in TLS, implementations MUST either abandon an association or rehandshake prior to allowing the sequence number to wrap.

Similarly, implementations MUST NOT allow the epoch to wrap, but instead MUST establish a new association, terminating the old association as described in Section 4.2.8. In practice, implementations rarely rehandshake repeatedly on the same channel, so this is not likely to be an issue.

4.1.1. Transport Layer Mapping

Each DTLS record MUST fit within a single datagram. In order to avoid IP fragmentation, clients of the DTLS record layer SHOULD attempt to size records so that they fit within any PMTU estimates obtained from the record layer.

Note that unlike IPsec, DTLS records do not contain any association identifiers. Applications must arrange to multiplex between associations. With UDP, this is presumably done with the host/port number.

Multiple DTLS records may be placed in a single datagram. They are simply encoded consecutively. The DTLS record framing is sufficient to determine the boundaries. Note, however, that the first byte of the datagram payload must be the beginning of a record. Records may not span datagrams.

Some transports, such as DCCP [DCCP] provide their own sequence numbers. When carried over those transports, both the DTLS and the transport sequence numbers will be present. Although this introduces a small amount of inefficiency, the transport layer and DTLS sequence numbers serve different purposes; therefore, for conceptual simplicity, it is superior to use both sequence numbers. In the future, extensions to DTLS may be specified that allow the use of only one set of sequence numbers for deployment in constrained environments.

Some transports, such as DCCP, provide congestion control for traffic carried over them. If the congestion window is sufficiently narrow, DTLS handshake retransmissions may be held rather than transmitted immediately, potentially leading to timeouts and spurious retransmission. When DTLS is used over such transports, care should be taken not to overrun the likely congestion window. [DCCPDTLS] defines a mapping of DTLS to DCCP that takes these issues into account.

4.1.1.1. PMTU Issues

In general, DTLS's philosophy is to leave PMTU discovery to the application. However, DTLS cannot completely ignore PMTU for three reasons:

- The DTLS record framing expands the datagram size, thus lowering the effective PMTU from the application's perspective.
- In some implementations, the application may not directly talk to the network, in which case the DTLS stack may absorb ICMP [RFC1191] "Datagram Too Big" indications or ICMPv6 [RFC4443] "Packet Too Big" indications.
- The DTLS handshake messages can exceed the PMTU.

In order to deal with the first two issues, the DTLS record layer SHOULD behave as described below.

If PMTU estimates are available from the underlying transport protocol, they should be made available to upper layer protocols. In particular:

- For DTLS over UDP, the upper layer protocol SHOULD be allowed to obtain the PMTU estimate maintained in the IP layer.
- For DTLS over DCCP, the upper layer protocol SHOULD be allowed to obtain the current estimate of the PMTU.
- For DTLS over TCP or SCTP, which automatically fragment and reassemble datagrams, there is no PMTU limitation. However, the upper layer protocol MUST NOT write any record that exceeds the maximum record size of 2^{14} bytes.

The DTLS record layer SHOULD allow the upper layer protocol to discover the amount of record expansion expected by the DTLS processing. Note that this number is only an estimate because of block padding and the potential use of DTLS compression.

If there is a transport protocol indication (either via ICMP or via a refusal to send the datagram as in Section 14 of [DCCP]), then the DTLS record layer MUST inform the upper layer protocol of the error.

The DTLS record layer SHOULD NOT interfere with upper layer protocols performing PMTU discovery, whether via [RFC1191] or [RFC4821] mechanisms. In particular:

- Where allowed by the underlying transport protocol, the upper layer protocol SHOULD be allowed to set the state of the DF bit (in IPv4) or prohibit local fragmentation (in IPv6).
- If the underlying transport protocol allows the application to request PMTU probing (e.g., DCCP), the DTLS record layer should honor this request.

The final issue is the DTLS handshake protocol. From the perspective of the DTLS record layer, this is merely another upper layer protocol. However, DTLS handshakes occur infrequently and involve only a few round trips; therefore, the handshake protocol PMTU handling places a premium on rapid completion over accurate PMTU discovery. In order to allow connections under these circumstances, DTLS implementations SHOULD follow the following rules:

- If the DTLS record layer informs the DTLS handshake layer that a message is too big, it SHOULD immediately attempt to fragment it, using any existing information about the PMTU.
- If repeated retransmissions do not result in a response, and the PMTU is unknown, subsequent retransmissions SHOULD back off to a smaller record size, fragmenting the handshake message as appropriate. This standard does not specify an exact number of retransmits to attempt before backing off, but 2-3 seems appropriate.

4.1.2. Record Payload Protection

Like TLS, DTLS transmits data as a series of protected records. The rest of this section describes the details of that format.

4.1.2.1. MAC

The DTLS MAC is the same as that of TLS 1.2. However, rather than using TLS's implicit sequence number, the sequence number used to compute the MAC is the 64-bit value formed by concatenating the epoch and the sequence number in the order they appear on the wire. Note that the DTLS epoch + sequence number is the same length as the TLS sequence number.

TLS MAC calculation is parameterized on the protocol version number, which, in the case of DTLS, is the on-the-wire version, i.e., {254, 253} for DTLS 1.2.

Note that one important difference between DTLS and TLS MAC handling is that in TLS, MAC errors must result in connection termination. In DTLS, the receiving implementation MAY simply discard the offending record and continue with the connection. This change is possible because DTLS records are not dependent on each other in the way that TLS records are.

In general, DTLS implementations SHOULD silently discard records with bad MACs or that are otherwise invalid. They MAY log an error. If a DTLS implementation chooses to generate an alert when it receives a message with an invalid MAC, it MUST generate a `bad_record_mac` alert

with level fatal and terminate its connection state. Note that because errors do not cause connection termination, DTLS stacks are more efficient error type oracles than TLS stacks. Thus, it is especially important that the advice in Section 6.2.3.2 of [TLS12] be followed.

4.1.2.2. Null or Standard Stream Cipher

The DTLS NULL cipher is performed exactly as the TLS 1.2 NULL cipher.

The only stream cipher described in TLS 1.2 is RC4, which cannot be randomly accessed. RC4 MUST NOT be used with DTLS.

4.1.2.3. Block Cipher

DTLS block cipher encryption and decryption are performed exactly as with TLS 1.2.

4.1.2.4. AEAD Ciphers

TLS 1.2 introduced authenticated encryption with additional data (AEAD) cipher suites. The existing AEAD cipher suites, defined in [ECCGCM] and [RSAGCM], can be used with DTLS exactly as with TLS 1.2.

4.1.2.5. New Cipher Suites

Upon registration, new TLS cipher suites MUST indicate whether they are suitable for DTLS usage and what, if any, adaptations must be made (see Section 7 for IANA considerations).

4.1.2.6. Anti-Replay

DTLS records contain a sequence number to provide replay protection. Sequence number verification SHOULD be performed using the following sliding window procedure, borrowed from Section 3.4.3 of [ESP].

The receiver packet counter for this session MUST be initialized to zero when the session is established. For each received record, the receiver MUST verify that the record contains a sequence number that does not duplicate the sequence number of any other record received during the life of this session. This SHOULD be the first check applied to a packet after it has been matched to a session, to speed rejection of duplicate records.

Duplicates are rejected through the use of a sliding receive window. (How the window is implemented is a local matter, but the following text describes the functionality that the implementation must exhibit.) A minimum window size of 32 MUST be supported, but a

window size of 64 is preferred and SHOULD be employed as the default. Another window size (larger than the minimum) MAY be chosen by the receiver. (The receiver does not notify the sender of the window size.)

The "right" edge of the window represents the highest validated sequence number value received on this session. Records that contain sequence numbers lower than the "left" edge of the window are rejected. Packets falling within the window are checked against a list of received packets within the window. An efficient means for performing this check, based on the use of a bit mask, is described in Section 3.4.3 of [ESP].

If the received record falls within the window and is new, or if the packet is to the right of the window, then the receiver proceeds to MAC verification. If the MAC validation fails, the receiver MUST discard the received record as invalid. The receive window is updated only if the MAC verification succeeds.

4.1.2.7. Handling Invalid Records

Unlike TLS, DTLS is resilient in the face of invalid records (e.g., invalid formatting, length, MAC, etc.). In general, invalid records SHOULD be silently discarded, thus preserving the association; however, an error MAY be logged for diagnostic purposes. Implementations which choose to generate an alert instead, MUST generate fatal level alerts to avoid attacks where the attacker repeatedly probes the implementation to see how it responds to various types of error. Note that if DTLS is run over UDP, then any implementation which does this will be extremely susceptible to denial-of-service (DoS) attacks because UDP forgery is so easy. Thus, this practice is NOT RECOMMENDED for such transports.

If DTLS is being carried over a transport that is resistant to forgery (e.g., SCTP with SCTP-AUTH), then it is safer to send alerts because an attacker will have difficulty forging a datagram that will not be rejected by the transport layer.

4.2. The DTLS Handshake Protocol

DTLS uses all of the same handshake messages and flows as TLS, with three principal changes:

1. A stateless cookie exchange has been added to prevent denial-of-service attacks.

2. Modifications to the handshake header to handle message loss, reordering, and DTLS message fragmentation (in order to avoid IP fragmentation).
3. Retransmission timers to handle message loss.

With these exceptions, the DTLS message formats, flows, and logic are the same as those of TLS 1.2.

4.2.1. Denial-of-Service Countermeasures

Datagram security protocols are extremely susceptible to a variety of DoS attacks. Two attacks are of particular concern:

1. An attacker can consume excessive resources on the server by transmitting a series of handshake initiation requests, causing the server to allocate state and potentially to perform expensive cryptographic operations.
2. An attacker can use the server as an amplifier by sending connection initiation messages with a forged source of the victim. The server then sends its next message (in DTLS, a Certificate message, which can be quite large) to the victim machine, thus flooding it.

In order to counter both of these attacks, DTLS borrows the stateless cookie technique used by Photuris [PHOTURIS] and IKE [IKEv2]. When the client sends its ClientHello message to the server, the server MAY respond with a HelloVerifyRequest message. This message contains a stateless cookie generated using the technique of [PHOTURIS]. The client MUST retransmit the ClientHello with the cookie added. The server then verifies the cookie and proceeds with the handshake only if it is valid. This mechanism forces the attacker/client to be able to receive the cookie, which makes DoS attacks with spoofed IP addresses difficult. This mechanism does not provide any defense against DoS attacks mounted from valid IP addresses.

The exchange is shown below:

```

Client                               Server
-----                             -
ClientHello                         ----->

                                   <----- HelloVerifyRequest
                                   (contains cookie)

ClientHello                         ----->
(with cookie)

[Rest of handshake]
```

DTLS therefore modifies the ClientHello message to add the cookie value.

```

struct {
    ProtocolVersion client_version;
    Random random;
    SessionID session_id;
    opaque cookie<0..2^8-1>; // New field
    CipherSuite cipher_suites<2..2^16-1>;
    CompressionMethod compression_methods<1..2^8-1>;
} ClientHello;
```

When sending the first ClientHello, the client does not have a cookie yet; in this case, the Cookie field is left empty (zero length).

The definition of HelloVerifyRequest is as follows:

```

struct {
    ProtocolVersion server_version;
    opaque cookie<0..2^8-1>;
} HelloVerifyRequest;
```

The HelloVerifyRequest message type is hello_verify_request(3).

The server_version field has the same syntax as in TLS. However, in order to avoid the requirement to do version negotiation in the initial handshake, DTLS 1.2 server implementations SHOULD use DTLS version 1.0 regardless of the version of TLS that is expected to be negotiated. DTLS 1.2 and 1.0 clients MUST use the version solely to indicate packet formatting (which is the same in both DTLS 1.2 and 1.0) and not as part of version negotiation. In particular, DTLS 1.2 clients MUST NOT assume that because the server uses version 1.0 in the HelloVerifyRequest that the server is not DTLS 1.2 or that it will eventually negotiate DTLS 1.0 rather than DTLS 1.2.

When responding to a HelloVerifyRequest, the client MUST use the same parameter values (version, random, session_id, cipher_suites, compression_method) as it did in the original ClientHello. The server SHOULD use those values to generate its cookie and verify that they are correct upon cookie receipt. The server MUST use the same version number in the HelloVerifyRequest that it would use when sending a ServerHello. Upon receipt of the ServerHello, the client MUST verify that the server version values match. In order to avoid sequence number duplication in case of multiple HelloVerifyRequests, the server MUST use the record sequence number in the ClientHello as the record sequence number in the HelloVerifyRequest.

Note: This specification increases the cookie size limit to 255 bytes for greater future flexibility. The limit remains 32 for previous versions of DTLS.

The DTLS server SHOULD generate cookies in such a way that they can be verified without retaining any per-client state on the server. One technique is to have a randomly generated secret and generate cookies as:

Cookie = HMAC(Secret, Client-IP, Client-Parameters)

When the second ClientHello is received, the server can verify that the Cookie is valid and that the client can receive packets at the given IP address. In order to avoid sequence number duplication in case of multiple cookie exchanges, the server MUST use the record sequence number in the ClientHello as the record sequence number in its initial ServerHello. Subsequent ServerHellos will only be sent after the server has created state and MUST increment normally.

One potential attack on this scheme is for the attacker to collect a number of cookies from different addresses and then reuse them to attack the server. The server can defend against this attack by changing the Secret value frequently, thus invalidating those cookies. If the server wishes that legitimate clients be able to handshake through the transition (e.g., they received a cookie with Secret 1 and then sent the second ClientHello after the server has changed to Secret 2), the server can have a limited window during which it accepts both secrets. [IKEv2] suggests adding a version number to cookies to detect this case. An alternative approach is simply to try verifying with both secrets.

DTLS servers SHOULD perform a cookie exchange whenever a new handshake is being performed. If the server is being operated in an environment where amplification is not a problem, the server MAY be configured not to perform a cookie exchange. The default SHOULD be that the exchange is performed, however. In addition, the server MAY

choose not to do a cookie exchange when a session is resumed. Clients MUST be prepared to do a cookie exchange with every handshake.

If HelloVerifyRequest is used, the initial ClientHello and HelloVerifyRequest are not included in the calculation of the handshake_messages (for the CertificateVerify message) and verify_data (for the Finished message).

If a server receives a ClientHello with an invalid cookie, it SHOULD treat it the same as a ClientHello with no cookie. This avoids race/deadlock conditions if the client somehow gets a bad cookie (e.g., because the server changes its cookie signing key).

Note to implementors: This may result in clients receiving multiple HelloVerifyRequest messages with different cookies. Clients SHOULD handle this by sending a new ClientHello with a cookie in response to the new HelloVerifyRequest.

4.2.2. Handshake Message Format

In order to support message loss, reordering, and message fragmentation, DTLS modifies the TLS 1.2 handshake header:

```
struct {
    HandshakeType msg_type;
    uint24 length;
    uint16 message_seq;           // New field
    uint24 fragment_offset;      // New field
    uint24 fragment_length;      // New field
    select (HandshakeType) {
        case hello_request: HelloRequest;
        case client_hello: ClientHello;
        case hello_verify_request: HelloVerifyRequest; // New type
        case server_hello: ServerHello;
        case certificate: Certificate;
        case server_key_exchange: ServerKeyExchange;
        case certificate_request: CertificateRequest;
        case server_hello_done: ServerHelloDone;
        case certificate_verify: CertificateVerify;
        case client_key_exchange: ClientKeyExchange;
        case finished: Finished;
    } body;
} Handshake;
```

The first message each side transmits in each handshake always has message_seq = 0. Whenever each new message is generated, the message_seq value is incremented by one. Note that in the case of a

rehandshake, this implies that the HelloRequest will have message_seq = 0 and the ServerHello will have message_seq = 1. When a message is retransmitted, the same message_seq value is used. For example:

```

Client                               Server
-----                               -
ClientHello (seq=0) ----->
                                X<-- HelloVerifyRequest (seq=0)
                                (lost)

[Timer Expires]

ClientHello (seq=0) ----->
(retransmit)

                                <----- HelloVerifyRequest (seq=0)

ClientHello (seq=1) ----->
(with cookie)

                                <----- ServerHello (seq=1)
                                <----- Certificate (seq=2)
                                <----- ServerHelloDone (seq=3)

[Rest of handshake]
```

Note, however, that from the perspective of the DTLS record layer, the retransmission is a new record. This record will have a new DTLSPlaintext.sequence_number value.

DTLS implementations maintain (at least notionally) a next_receive_seq counter. This counter is initially set to zero. When a message is received, if its sequence number matches next_receive_seq, next_receive_seq is incremented and the message is processed. If the sequence number is less than next_receive_seq, the message MUST be discarded. If the sequence number is greater than next_receive_seq, the implementation SHOULD queue the message but MAY discard it. (This is a simple space/bandwidth tradeoff).

4.2.3. Handshake Message Fragmentation and Reassembly

As noted in Section 4.1.1, each DTLS message MUST fit within a single transport layer datagram. However, handshake messages are potentially bigger than the maximum record size. Therefore, DTLS provides a mechanism for fragmenting a handshake message over a number of records, each of which can be transmitted separately, thus avoiding IP fragmentation.

When transmitting the handshake message, the sender divides the message into a series of N contiguous data ranges. These ranges MUST NOT be larger than the maximum handshake fragment size and MUST jointly contain the entire handshake message. The ranges SHOULD NOT overlap. The sender then creates N handshake messages, all with the same message_seq value as the original handshake message. Each new message is labeled with the fragment_offset (the number of bytes contained in previous fragments) and the fragment_length (the length of this fragment). The length field in all messages is the same as the length field of the original message. An unfragmented message is a degenerate case with fragment_offset=0 and fragment_length=length.

When a DTLS implementation receives a handshake message fragment, it MUST buffer it until it has the entire handshake message. DTLS implementations MUST be able to handle overlapping fragment ranges. This allows senders to retransmit handshake messages with smaller fragment sizes if the PMTU estimate changes.

Note that as with TLS, multiple handshake messages may be placed in the same DTLS record, provided that there is room and that they are part of the same flight. Thus, there are two acceptable ways to pack two DTLS messages into the same datagram: in the same record or in separate records.

4.2.4. Timeout and Retransmission

DTLS messages are grouped into a series of message flights, according to the diagrams below. Although each flight of messages may consist of a number of messages, they should be viewed as monolithic for the purpose of timeout and retransmission.

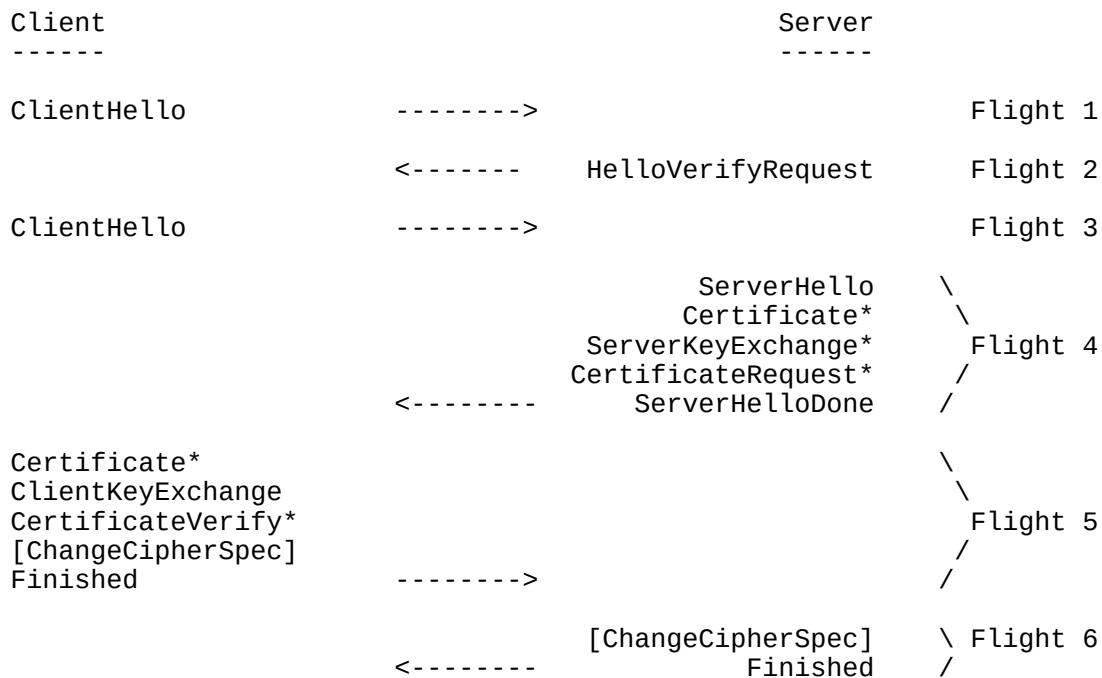
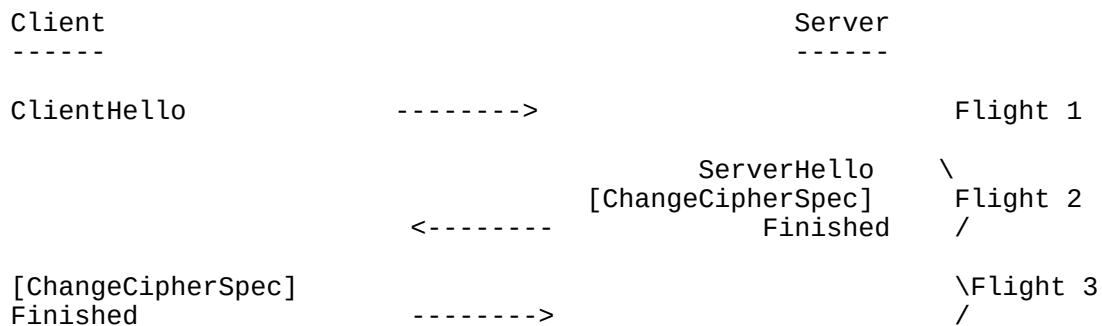


Figure 1. Message Flights for Full Handshake

Figure 2. Message Flights for Session-Resuming Handshake
(No Cookie Exchange)

DTLS uses a simple timeout and retransmission scheme with the following state machine. Because DTLS clients send the first message (ClientHello), they start in the PREPARING state. DTLS servers start in the WAITING state, but with empty buffers and no retransmit timer.

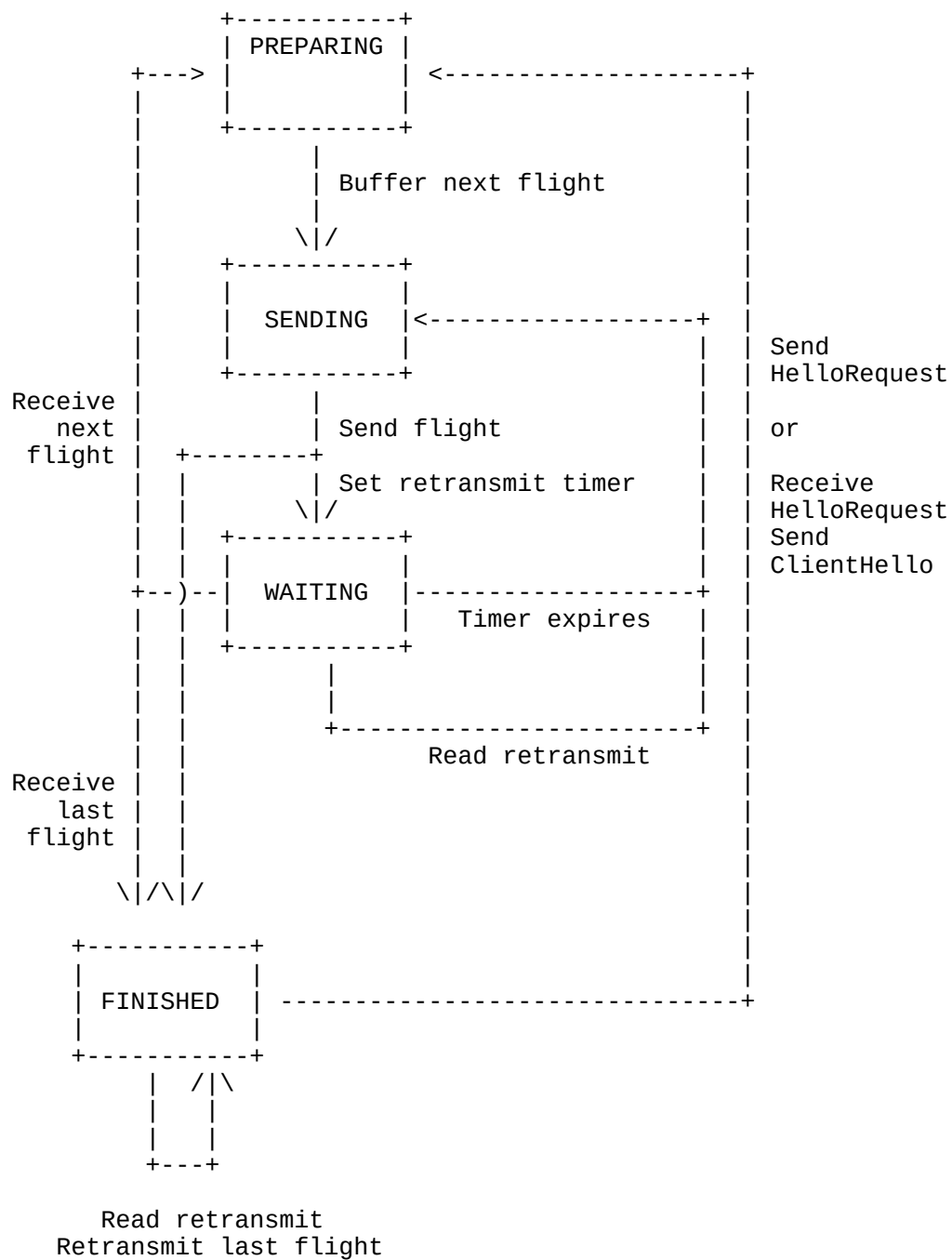


Figure 3. DTLS Timeout and Retransmission State Machine

The state machine has three basic states.

In the PREPARING state, the implementation does whatever computations are necessary to prepare the next flight of messages. It then buffers them up for transmission (emptying the buffer first) and enters the SENDING state.

In the SENDING state, the implementation transmits the buffered flight of messages. Once the messages have been sent, the implementation then enters the FINISHED state if this is the last flight in the handshake. Or, if the implementation expects to receive more messages, it sets a retransmit timer and then enters the WAITING state.

There are three ways to exit the WAITING state:

1. The retransmit timer expires: the implementation transitions to the SENDING state, where it retransmits the flight, resets the retransmit timer, and returns to the WAITING state.
2. The implementation reads a retransmitted flight from the peer: the implementation transitions to the SENDING state, where it retransmits the flight, resets the retransmit timer, and returns to the WAITING state. The rationale here is that the receipt of a duplicate message is the likely result of timer expiry on the peer and therefore suggests that part of one's previous flight was lost.
3. The implementation receives the next flight of messages: if this is the final flight of messages, the implementation transitions to FINISHED. If the implementation needs to send a new flight, it transitions to the PREPARING state. Partial reads (whether partial messages or only some of the messages in the flight) do not cause state transitions or timer resets.

Because DTLS clients send the first message (ClientHello), they start in the PREPARING state. DTLS servers start in the WAITING state, but with empty buffers and no retransmit timer.

When the server desires a rehandshake, it transitions from the FINISHED state to the PREPARING state to transmit the HelloRequest. When the client receives a HelloRequest, it transitions from FINISHED to PREPARING to transmit the ClientHello.

In addition, for at least twice the default MSL defined for [TCP], when in the FINISHED state, the node that transmits the last flight (the server in an ordinary handshake or the client in a resumed handshake) MUST respond to a retransmit of the peer's last flight

with a retransmit of the last flight. This avoids deadlock conditions if the last flight gets lost. This requirement applies to DTLS 1.0 as well, and though not explicit in [DTLS1], it was always required for the state machine to function correctly. To see why this is necessary, consider what happens in an ordinary handshake if the server's Finished message is lost: the server believes the handshake is complete but it actually is not. As the client is waiting for the Finished message, the client's retransmit timer will fire and it will retransmit the client's Finished message. This will cause the server to respond with its own Finished message, completing the handshake. The same logic applies on the server side for the resumed handshake.

Note that because of packet loss, it is possible for one side to be sending application data even though the other side has not received the first side's Finished message. Implementations **MUST** either discard or buffer all application data packets for the new epoch until they have received the Finished message for that epoch. Implementations **MAY** treat receipt of application data with a new epoch prior to receipt of the corresponding Finished message as evidence of reordering or packet loss and retransmit their final flight immediately, shortcutting the retransmission timer.

4.2.4.1. Timer Values

Though timer values are the choice of the implementation, mishandling of the timer can lead to serious congestion problems; for example, if many instances of a DTLS time out early and retransmit too quickly on a congested link. Implementations **SHOULD** use an initial timer value of 1 second (the minimum defined in RFC 6298 [RFC6298]) and double the value at each retransmission, up to no less than the RFC 6298 maximum of 60 seconds. Note that we recommend a 1-second timer rather than the 3-second RFC 6298 default in order to improve latency for time-sensitive applications. Because DTLS only uses retransmission for handshake and not dataflow, the effect on congestion should be minimal.

Implementations **SHOULD** retain the current timer value until a transmission without loss occurs, at which time the value may be reset to the initial value. After a long period of idleness, no less than 10 times the current timer value, implementations may reset the timer to the initial value. One situation where this might occur is when a rehandshake is used after substantial data transfer.

4.2.5. ChangeCipherSpec

As with TLS, the ChangeCipherSpec message is not technically a handshake message but **MUST** be treated as part of the same flight as the associated Finished message for the purposes of timeout and retransmission. This creates a potential ambiguity because the order of the ChangeCipherSpec cannot be established unambiguously with respect to the handshake messages in case of message loss.

This is not a problem with any current TLS mode because the expected set of handshake messages logically preceeding the ChangeCipherSpec is predictable from the rest of the handshake state. However, future modes **MUST** take care to avoid creating ambiguity.

4.2.6. CertificateVerify and Finished Messages

CertificateVerify and Finished messages have the same format as in TLS. Hash calculations include entire handshake messages, including DTLS-specific fields: message_seq, fragment_offset, and fragment_length. However, in order to remove sensitivity to handshake message fragmentation, the Finished MAC **MUST** be computed as if each handshake message had been sent as a single fragment. Note that in cases where the cookie exchange is used, the initial ClientHello and HelloVerifyRequest **MUST NOT** be included in the CertificateVerify or Finished MAC computations.

4.2.7. Alert Messages

Note that Alert messages are not retransmitted at all, even when they occur in the context of a handshake. However, a DTLS implementation which would ordinarily issue an alert **SHOULD** generate a new alert message if the offending record is received again (e.g., as a retransmitted handshake message). Implementations **SHOULD** detect when a peer is persistently sending bad messages and terminate the local connection state after such misbehavior is detected.

4.2.8. Establishing New Associations with Existing Parameters

If a DTLS client-server pair is configured in such a way that repeated connections happen on the same host/port quartet, then it is possible that a client will silently abandon one connection and then initiate another with the same parameters (e.g., after a reboot). This will appear to the server as a new handshake with epoch=0. In cases where a server believes it has an existing association on a given host/port quartet and it receives an epoch=0 ClientHello, it **SHOULD** proceed with a new handshake but **MUST NOT** destroy the existing association until the client has demonstrated reachability either by completing a cookie exchange or by completing a complete handshake.

including delivering a verifiable Finished message. After a correct Finished message is received, the server MUST abandon the previous association to avoid confusion between two valid associations with overlapping epochs. The reachability requirement prevents off-path/blind attackers from destroying associations merely by sending forged ClientHellos.

4.3. Summary of New Syntax

This section includes specifications for the data structures that have changed between TLS 1.2 and DTLS 1.2. See [TLS12] for the definition of this syntax.

4.3.1. Record Layer

```

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 epoch;                // New field
    uint48 sequence_number;     // New field
    uint16 length;
    opaque fragment[DTLSPplaintext.length];
} DTLSPplaintext;

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 epoch;                // New field
    uint48 sequence_number;     // New field
    uint16 length;
    opaque fragment[DTLSCompressed.length];
} DTLSCompressed;

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 epoch;                // New field
    uint48 sequence_number;     // New field
    uint16 length;
    select (CipherSpec.cipher_type) {
        case block: GenericBlockCipher;
        case aead:  GenericAEADCipher; // New field
    } fragment;
} DTLSCiphertext;
```

4.3.2. Handshake Protocol

```

enum {
    hello_request(0), client_hello(1), server_hello(2),
    hello_verify_request(3), // New field
    certificate(11), server_key_exchange(12),
    certificate_request(13), server_hello_done(14),
    certificate_verify(15), client_key_exchange(16),
    finished(20), (255) } HandshakeType;

struct {
    HandshakeType msg_type;
    uint24 length;
    uint16 message_seq; // New field
    uint24 fragment_offset; // New field
    uint24 fragment_length; // New field
    select (HandshakeType) {
        case hello_request: HelloRequest;
        case client_hello: ClientHello;
        case server_hello: ServerHello;
        case hello_verify_request: HelloVerifyRequest; // New field
        case certificate: Certificate;
        case server_key_exchange: ServerKeyExchange;
        case certificate_request: CertificateRequest;
        case server_hello_done: ServerHelloDone;
        case certificate_verify: CertificateVerify;
        case client_key_exchange: ClientKeyExchange;
        case finished: Finished;
    } body; } Handshake;

struct {
    ProtocolVersion client_version;
    Random random;
    SessionID session_id;
    opaque cookie<0..2^8-1>; // New field
    CipherSuite cipher_suites<2..2^16-1>;
    CompressionMethod compression_methods<1..2^8-1>; } ClientHello;

struct {
    ProtocolVersion server_version;
    opaque cookie<0..2^8-1>; } HelloVerifyRequest;

```

5. Security Considerations

This document describes a variant of TLS 1.2; therefore, most of the security considerations are the same as those of TLS 1.2 [TLS12], described in Appendices D, E, and F.

The primary additional security consideration raised by DTLS is that of denial of service. DTLS includes a cookie exchange designed to protect against denial of service. However, implementations that do not use this cookie exchange are still vulnerable to DoS. In particular, DTLS servers that do not use the cookie exchange may be used as attack amplifiers even if they themselves are not experiencing DoS. Therefore, DTLS servers SHOULD use the cookie exchange unless there is good reason to believe that amplification is not a threat in their environment. Clients MUST be prepared to do a cookie exchange with every handshake.

Unlike TLS implementations, DTLS implementations SHOULD NOT respond to invalid records by terminating the connection. See Section 4.1.2.7 for details on this.

6. Acknowledgments

The authors would like to thank Dan Boneh, Eu-Jin Goh, Russ Housley, Constantine Sapuntzakis, and Hovav Shacham for discussions and comments on the design of DTLS. Thanks to the anonymous NDSS reviewers of our original NDSS paper on DTLS [DTLS] for their comments. Also, thanks to Steve Kent for feedback that helped clarify many points. The section on PMTU was cribbed from the DCCP specification [DCCP]. Pasi Eronen provided a detailed review of this specification. Peter Saint-Andre provided the list of changes in Section 8. Helpful comments on the document were also received from Mark Allman, Jari Arkko, Mohamed Badra, Michael D'Errico, Adrian Farrell, Joel Halpern, Ted Hardie, Charlia Kaufman, Pekka Savola, Allison Mankin, Nikos Mavrogiannopoulos, Alexey Melnikov, Robin Seggelmann, Michael Tuexen, Juho Vaha-Herttua, and Florian Weimer.

7. IANA Considerations

This document uses the same identifier space as TLS [TLS12], so no new IANA registries are required. When new identifiers are assigned for TLS, authors MUST specify whether they are suitable for DTLS. IANA has modified all TLS parameter registries to add a DTLS-OK flag, indicating whether the specification may be used with DTLS. At the time of publication, all of the [TLS12] registrations except the following are suitable for DTLS. The full table of registrations is available at [IANA].

From the TLS Cipher Suite Registry:

0x00,0x03	TLS_RSA_EXPORT_WITH_RC4_40_MD5	[RFC4346]
0x00,0x04	TLS_RSA_WITH_RC4_128_MD5	[RFC5246]
0x00,0x05	TLS_RSA_WITH_RC4_128_SHA	[RFC5246]
0x00,0x17	TLS_DH_anon_EXPORT_WITH_RC4_40_MD5	[RFC4346]

0x00,0x18	TLS_DH_anon_WITH_RC4_128_MD5	[RFC5246]
0x00,0x20	TLS_KRB5_WITH_RC4_128_SHA	[RFC2712]
0x00,0x24	TLS_KRB5_WITH_RC4_128_MD5	[RFC2712]
0x00,0x28	TLS_KRB5_EXPORT_WITH_RC4_40_SHA	[RFC2712]
0x00,0x2B	TLS_KRB5_EXPORT_WITH_RC4_40_MD5	[RFC2712]
0x00,0x8A	TLS_PSK_WITH_RC4_128_SHA	[RFC4279]
0x00,0x8E	TLS_DHE_PSK_WITH_RC4_128_SHA	[RFC4279]
0x00,0x92	TLS_RSA_PSK_WITH_RC4_128_SHA	[RFC4279]
0xC0,0x02	TLS_ECDH_ECDSA_WITH_RC4_128_SHA	[RFC4492]
0xC0,0x07	TLS_ECDHE_ECDSA_WITH_RC4_128_SHA	[RFC4492]
0xC0,0x0C	TLS_ECDH_RSA_WITH_RC4_128_SHA	[RFC4492]
0xC0,0x11	TLS_ECDHE_RSA_WITH_RC4_128_SHA	[RFC4492]
0xC0,0x16	TLS_ECDH_anon_WITH_RC4_128_SHA	[RFC4492]
0xC0,0x33	TLS_ECDHE_PSK_WITH_RC4_128_SHA	[RFC5489]

From the TLS Exporter Label Registry:

client	EAP encryption	[RFC5216]
ttls	keying material	[RFC5281]
ttls	challenge	[RFC5281]

This document defines a new handshake message, `hello_verify_request`, whose value has been allocated from the TLS HandshakeType registry defined in [TLS12]. The value "3" has been assigned by the IANA.

8. Changes since DTLS 1.0

This document reflects the following changes since DTLS 1.0 [DTLS1].

- Updated to match TLS 1.2 [TLS12].
- Addition of AEAD Ciphers in Section 4.1.2.3 (tracking changes in TLS 1.2).
- Clarifications regarding sequence numbers and epochs in Section 4.1 and a clear procedure for dealing with state loss in Section 4.2.8.
- Clarifications and more detailed rules regarding Path MTU issues in Section 4.1.1.1. Clarification of the fragmentation text throughout.
- Clarifications regarding handling of invalid records in Section 4.1.2.7.
- A new paragraph describing handling of invalid cookies at the end of Section 4.2.1.

- Some new text describing how to avoid handshake deadlock conditions at the end of Section 4.2.4.
- Some new text about CertificateVerify messages in Section 4.2.6.
- A prohibition on epoch wrapping in Section 4.1.
- Clarification of the IANA requirements and the explicit requirement for a new IANA registration flag for each parameter.
- Added a record sequence number mirroring technique for handling repeated ClientHello messages.
- Recommend a fixed version number for HelloVerifyRequest.
- Numerous editorial changes.

9. References

9.1. Normative References

- [REQ] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997.
- [RFC1191] Mogul, J. and S. Deering, "Path MTU discovery", RFC 1191, November 1990.
- [RFC4301] Kent, S. and K. Seo, "Security Architecture for the Internet Protocol", RFC 4301, December 2005.
- [RFC4443] Conta, A., Deering, S., and M. Gupta, Ed., "Internet Control Message Protocol (ICMPv6) for the Internet Protocol Version 6 (IPv6) Specification", RFC 4443, March 2006.
- [RFC4821] Mathis, M. and J. Heffner, "Packetization Layer Path MTU Discovery", RFC 4821, March 2007.
- [RFC6298] Paxson, V., Allman, M., Chu, J., and M. Sargent, "Computing TCP's Retransmission Timer", RFC 6298, June 2011.
- [RSAGCM] Salowey, J., Choudhury, A., and D. McGrew, "AES Galois Counter Mode (GCM) Cipher Suites for TLS", RFC 5288, August 2008.
- [TCP] Postel, J., "Transmission Control Protocol", STD 7, RFC 793, September 1981.

- [TLS12] Dierks, T. and E. Rescorla, "The Transport Layer Security (TLS) Protocol Version 1.2", RFC 5246, August 2008.

9.2. Informative References

- [DCCP] Kohler, E., Handley, M., and S. Floyd, "Datagram Congestion Control Protocol (DCCP)", RFC 4340, March 2006.
- [DCCPDTLS] Phelan, T., "Datagram Transport Layer Security (DTLS) over the Datagram Congestion Control Protocol (DCCP)", RFC 5238, May 2008.
- [DTLS] Modadugu, N. and E. Rescorla, "The Design and Implementation of Datagram TLS", Proceedings of ISOC NDSS 2004, February 2004.
- [DTLS1] Rescorla, E. and N. Modadugu, "Datagram Transport Layer Security", RFC 4347, April 2006.
- [ECCGCM] Rescorla, E., "TLS Elliptic Curve Cipher Suites with SHA-256/384 and AES Galois Counter Mode (GCM)", RFC 5289, August 2008.
- [ESP] Kent, S., "IP Encapsulating Security Payload (ESP)", RFC 4303, December 2005.
- [IANA] IANA, "Transport Layer Security (TLS) Parameters", <http://www.iana.org/assignments/tls-parameters>.
- [IKEv2] Kaufman, C., Hoffman, P., Nir, Y., and P. Eronen, "Internet Key Exchange Protocol Version 2 (IKEv2)", RFC 5996, September 2010.
- [IMAP] Crispin, M., "INTERNET MESSAGE ACCESS PROTOCOL - VERSION 4rev1", RFC 3501, March 2003.
- [PHOTURIS] Karn, P. and W. Simpson, "Photuris: Session-Key Management Protocol", RFC 2522, March 1999.
- [POP] Myers, J. and M. Rose, "Post Office Protocol - Version 3", STD 53, RFC 1939, May 1996.
- [SIP] Rosenberg, J., Schulzrinne, H., Camarillo, G., Johnston, A., Peterson, J., Sparks, R., Handley, M., and E. Schooler, "SIP: Session Initiation Protocol", RFC 3261, June 2002.

- [TLS] Dierks, T. and C. Allen, "The TLS Protocol Version 1.0", RFC 2246, January 1999.
- [TLS11] Dierks, T. and E. Rescorla, "The Transport Layer Security (TLS) Protocol Version 1.1", RFC 4346, April 2006.
- [WHYIPSEC] Bellovin, S., "Guidelines for Specifying the Use of IPsec Version 2", BCP 146, RFC 5406, February 2009.

Authors' Addresses

Eric Rescorla
RTFM, Inc.
2064 Edgewood Drive
Palo Alto, CA 94303

Email: ekr@rtfm.com

Nagendra Modadugu
Google, Inc.

Email: nagendra@cs.stanford.edu